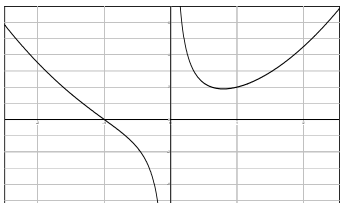
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 5(d) | $\frac{20}{x}+\frac{10}{2x+1}=45$ oe | **M2** | **B1** for $\frac{20}{x}$ seen or $\frac{10}{2x+1}$ seen |
| | $90x^2 - 5x - 20$ [= 0] oe | **B2** | **B1** for $\frac{20(2x+1)+10x}{x(2x+1)}$ = 45 or better |
| | $(9x + 4)(2x - 1)$ [= 0] or for $\frac{--1\pm\sqrt{(-1)^2-4(18)(-4)}}{2(18)}$ oe | **M2** | **FT** *their* 3-term quadratic<br>**M1** for factors that give two correct terms when expanded<br>or for correct discriminant or correct $\frac{-b}{2a}$ provided quadratic formula is in correct form |
| | [0].5 or $\frac{1}{2}$ final answer | **B1** | |
| 6(a)(i) | $\frac{1}{3}$ oe | **1** | |
| 6(a)(ii) | 0 | **1** | |
| 6(a)(iii) | $\frac{1}{6}$ oe | **1** | |
| 6(b)(i) | $\frac{1}{15}$ oe | **2** | **M1** for $\frac{2}{6}\times\frac{1}{5}$ or equivalent method |
| 6(b)(ii) | $\frac{4}{15}$ oe | **3** | **M2** for $\frac{2}{6}\times\frac{1}{5}+\frac{3}{6}\times\frac{2}{5}$ or equivalent method<br>or **M1** for $\frac{2}{6}\times\frac{1}{5}$ oe seen or $\frac{3}{6}\times\frac{2}{5}$ oe seen |
| 6(c) | $\frac{7}{18}$ oe | **3** | **M2** for $\left(\frac{1}{6}\right)^2+\left(\frac{2}{6}\right)^2+\left(\frac{3}{6}\right)^2$ oe<br>or **M1** for one correct product seen<br>or sample space with 14 correct pairs identified |
| 7(a) | 2, 4.5 | **2** | **B1** for each |
| 7(b) | Correct graph<br>  | **4** | **B3 FT** for 6 or 7 correct points FT *their* table<br>or **B2 FT** for 4 or 5 correct points FT *their* table<br>or **B1 FT** for 2 or 3 correct points FT *their* table |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(c)(i) | –0.5 to –0.4 | **1** | |
| 7(c)(ii) | $y = 1 - x$ ruled<br>**and**<br>–1.9 to –1.75 | **2** | **M1** for [$y$ =] $1 - x$ or $\left[x^2 + \frac{1}{x} =\right] 1 - x$ soi<br>or **B1** for –1.9 to –1.75 |
| 7(d) | Any integer $\geq 2$ | **1** | |
| 8(a) | [$v$ = ] 40<br>[$w$ = ] 80<br>[$x$ = ] 40<br>[$y$ = ] 100<br>[$z$ = ] 60 | **5** | **B1** for each<br>**FT** angle $z$ as 140 – *their w* |
| 8(b) | 24 | **3** | **M2** for $360 - 11x = 2 \times 2x$ oe<br>or **M1** for $360 - 11x$ seen<br>or obtuse angle $KOL = 2 \times 2x$ oe |
| 8(c)(i) | angle $ADX$ = angle $BCX$ oe<br>same segment oe<br><br>angle $DAX$ = angle $CBX$ oe<br>same segment oe<br><br>angle $AXD = BXC$ oe<br>[vertically] opposite oe | **M2** | Accept in any order<br>**M1** for one correct pair with reason<br><br>If 0 scored, **SC1** for two correct pairs of equal angles identified with incorrect/no reasons |
| | corresponding angles are equal oe | **A1** | |
| 8(c)(ii)(a) | 8.75 or 8¾ | **2** | **M1** for $\frac{8}{10} = \frac{7}{DX}$ oe |
| 8(c)(ii)(b) | 81.8 or 81.78 to 81.79 | **4** | **M2** for [cos[$BXC$] =] $\frac{5^2 + 7^2 - 8^2}{2 \times 5 \times 7}$ oe<br>or **M1** for $8^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \cos(...)$ oe<br>**A1** for $\frac{10}{70}$ oe |